

Rui Zeng

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🏛 Shenzhen Technology University · 🎓 Computer Science and Technology

Fields of Interest: Robotics, Embodied AI, Intelligent Systems, Human-Robot Interaction, Sim-to-Real Robotics

🎓 Education

Sep. 2023 | **Shenzhen Technology University** · School of Artificial Intelligence · Computer Science and Technology · B.Eng.
Present

⚙️ Core Competencies

- **Robotics and Embodied AI Practice:** Experienced in developing robotic systems that integrate perception, control, planning, and real-world task execution.
- **Sim-to-Real Deployment:** Participated in multiple robotics projects involving simulation training, physical robot deployment, system debugging, and performance improvement in real-world environments.
- **Human-Robot Interaction and Intelligent Systems:** Built AI-driven interactive systems involving LLM/VLM integration, voice interaction, edge AI deployment, and device-cloud communication.
- **Hardware-Software Integration:** Experienced in embedded development, motor control, sensor integration, camera-based perception, SLAM navigation, and robotic manipulation.
- **Team Collaboration and Product-Oriented Engineering:** Worked in interdisciplinary teams and contributed to projects ranging from research prototypes to batch-produced educational robot products.

🏆 Awards

RoboMaster University Sim2Real Challenge	Second Prize	May 2024
ICRA 2024 AgileX Sim2Real Challenge	First Prize	May 2024
ICRA 2025 Sim2Real Challenge	Second Prize	Jun. 2025
2025 RoboMaster University Championship	Second Prize	Aug. 2025
2025 RoboMaster University Championship Engineering Robot Group	Third Prize	Aug. 2025
China Computer Federation 2024 Tech Marathon	Third Prize	Oct. 2024
RAICOM Treasure Hunt	Third Prize	Aug. 2024

🔗 Project Experience

Humanoid Whole-Body Control

Project Description: Participated in a humanoid whole-body control project that converts human motion videos into robot motion data and trains humanoid robots to reproduce human movements through motion retargeting and reinforcement learning.

Selected Contributions & Outcomes:

- Contributed to an humanoid motion tracking pipeline, connecting human skeletal motion extraction, robot retargeting, policy training, simulation validation, and real-world humanoid execution.
- Explored robust humanoid motion tracking under imperfect human motion data and helped build

demonstrations connecting human motion input, simulation tracking, and real robot deployment.

- › Co-first author of a related research paper submitted to **NeurIPS 2026** and currently under review..

ICRA 2024 Sim2Real Challenge

Project Description: Developed an integrated perception and control system for a mobile manipulator, combining SLAM-based navigation, object detection, grasp pose prediction, and robotic manipulation for autonomous task execution in both simulation and real-world environments.

Selected Contributions & Outcomes:

- › Contributed to SLAM-based localization, autonomous navigation, robotic arm inverse kinematics, and manipulation module integration.
- › Participated in grasp pose prediction integration, task-level execution, and sim-to-real transfer of the system to a physical robot.
- › Successfully supported real-world autonomous manipulation tasks and won **First Prize / 2nd Place** in the ICRA 2024 Sim2Real Challenge.

STEM Education Robot Kit

Project Description: Collaborated on the development of a STEM educational robot kit integrating camera perception, AprilTag localization, path planning visualization, pallet pickup, and placement functions for hands-on robotics education.

Selected Contributions & Outcomes:

- › Participated in hardware-software integration, including camera-based perception, AprilTag localization and alignment, robot pose visualization, and planned path display.
- › Helped implement pallet pickup and placement tasks, and supported robot assembly, testing, iteration, and user-oriented improvement.
- › Helped transform the robot from a prototype into a **batch-produced educational product** later deployed and sold to student users.

Cathay Hackathon – AI Companion for Unaccompanied Children

Project Description: Built an edge-AI companion prototype for unaccompanied children during air travel, supporting multimodal sensing, voice interaction, AI dialogue, and device-cloud communication.

Selected Contributions & Outcomes:

- › Focused on the edge AI architecture and LLM interaction pipeline, including online/offline voice interaction workflow design.
- › Integrated multimodal sensing modules such as attitude sensing and non-contact biometric sensing, and supported device-to-parent communication.
- › Built a functional prototype within a hackathon setting and applied LLM-based interaction to a real human-centered scenario.